PNAS RESEARCH ARTICLE | CHEMISTRY

# Broadly applicable hydrophilic additive enhances electrochemical transistor function

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Contributed by Tobin J. Marks; received September 22, 2025; accepted January 23, 2026; reviewed by David Ginger, Howard E. Katz, and Thuc-Quyen Nguyen

**Organic semiconductors offering efficient mixed ionic-electronic charge transport are key components of organic electrochemical transistors (OECTs) needed for future bioelectronics and other technologies. However, hydrophobic semiconductors typically have limited ion mobility and are unstable in aqueous environments, restricting OECT applications. To address these issues, we report a broadly applicable strategy for high-performance OECTs by blending polymeric semiconductors with a photocrosslinkable hydrophilic ion-conducting supplement, poly(ethyleneglycol)-dimethylacrylate (PEGDMA). The result is ordered, interconnected semiconductor domains within an amorphous ion-conducting matrix, enabling rapid and reversible doping/dedoping without compromising charge transport. This approach enhances OECT performance across diverse electrolytes, semiconductors, and device architectures. Furthermore, PEGDMA enables high-resolution photopatterning of both semiconductors (<0.4 μm) and electrolytes (<2 μm), affording high-stability OECTs sustaining >10,000 cycles. This approach also enables wafer-scale array fabrication of 2,548 OECTs on 2" wafer, and miniaturized inverter, NAND, and NOR circuits. Also demonstrated are integration of these OECTs with a photosensor, creating a vision sensing array (10 × 10 pixels) that mimics visual image processing similar to the brain's perception system.**

hydrophobic semiconductor | hydrophilic additive | organic transistor | organic electrochemical transistor

## Significance

Efficient organic electrochemical transistor (OECT) operation requires semiconductors with crystalline domains for charge transport and amorphous regions for reversible ion exchange. However, widely used P/N-type organic semiconductors are hydrophobic, thus face significant challenges in mixed ionic-electronic conduction. Here, we introduce a simple, scalable, and efficient approach via blending a commercial hydrophilic additive, poly(ethylene glycol) dimethyl acrylate (PEGDMA) into semiconductors, achieving significant advancements in electrochemical transistor materials, devices, and systems. PEGDMA establishes percolated ion pathways without compromising electronic transport, broadens electrolyte and device compatibility, and enables high-resolution photopatterning of both semiconductors and electrolytes. The resulting OECTs operate stably in liquid and solid states, supporting wafer-scale integration, high-yield fabrication, reproducible, robust logic circuits, and a vision-sensor array that emulates human retinal compression.

Organic electrochemical transistors (OECTs) have great potential in areas such as intelligent sensing, neural recording, neuromorphic computation, and human–machine interfaces (1–18). Efficient OECT operation requires semiconducting materials with both ordered/crystalline regions for sizeable electron or hole transport and amorphous regions to facilitate reversible ion transport throughout the bulk material (6, 19, 20). Recent research has focused on designing amphiphilic organic mixed ionic-electronic conductors (OMIECs), which combine high-mobility π-conjugated backbones (e.g., polythiophene, diketopyrrolopyrrole, naphthalenediimide) with hydrophilic side chains (e.g., hydroxyl, lysine, carboxylic, oligoether). These side chains enhance hydration and ion penetration, promoting efficient volumetric electrochemical doping/dedoping (21–25). However, many OMIECs are derived from hydrophobic semiconductors originally developed for organic field-effect transistors (OFETs) and organic photovoltaics (OPVs). Thus, integrating hydrophilic side chains into these materials is synthetically challenging, synthetically laborious, and frequently incurs poor overall yields as well as poor OMIEC performance—high OFET or OPV performance does not always translate to that in OECTs (10, 26). Hydrophobic semiconductors, whether electron-transporting (n-type) or hole-transporting (p-type), often struggle to achieve mixed ionic-electronic conduction due to complex structural/electronic interactions and reaction with $O_2$ and moisture (19). Furthermore, optimizing OMIECs requires careful tuning of molecular structure and functionality, such as conjugated backbones and side chains, to control properties such as ion permeability, redox stability, and operational voltage (27–31). Therefore, exclusively focusing on new OMIEC functional materials may be costly and limit the broader application of traditional organic semiconductors.

To address these issues, specific materials and device engineering approaches, such as introducing porous and/or honeycomb semiconductor film morphologies and integrating ion exchange gels (32–40), have been reported to leverage the extensive library of high-performance and commercially available hydrophobic semiconductors. Despite increased electrolyte-semiconductor interfacial area and ion transport pathways, which

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This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2523877123/-/DCSupplemental.

Published March 2, 2026.

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enhance transconductance and operating frequencies, major challenges remain: i. These approaches are primarily applicable to p-type semiconductors, and are not effective for n-type or small-molecule hydrophobic semiconductors (41, 42). ii. The differences in hydrophilicity between semiconductors and electrolytes requires careful selection of ionic components based on their insertion capabilities relative to the channel materials (43). iii. In several cases operation mostly relies on an electrical double-layer mechanism rather than fully volumetric Faradaic electrochemical doping (5). iv. Current fabrication techniques are limited in scalability for OECT miniaturization and dense integration due to the constraints in scalable patterning methods for both semiconductors and electrolytes (44). Introduction of a vertical OECT (vOECT) architecture expands the versatility of OECTs with advantages such as high current densities, stable switching characteristics, and increased integration density (45, 46). In the first-generation vOECTs a redox-inactive cellulosic additive, Cin-Cell, was used to stabilize the vertical architecture during operation, however, it invariably reduces the electroactive component portion, thus depressing performance (42). Furthermore, vOECTs are even more sensitive than conventional OECTs (cOECTs) to the semiconductor hydrophilicity since ion exchange with the electrolyte occurs via nanoscopic orifices vs. the much larger voids in cOECT architectures. Vertical internal-gated OECTs also require the integration with ion reservoirs within the ion-permeable channel materials (e.g., PEDOT:PSS) to establish localized electrochemical doping sites for high-frequency operation (47, 48). Consequently, the limitations of ion-impermeable hydrophobic semiconductors in vOECTs are amplified, leading to lower performance (49–52). Therefore, to advance OECT development across all key aspects—materials applicability, scalable fabrication, device architectures, circuit/system integration, and application, it is essential to explore generally applicable efficient molecular designs and device engineering strategies for diverse semiconductors to realize families of high-performing OECTs.

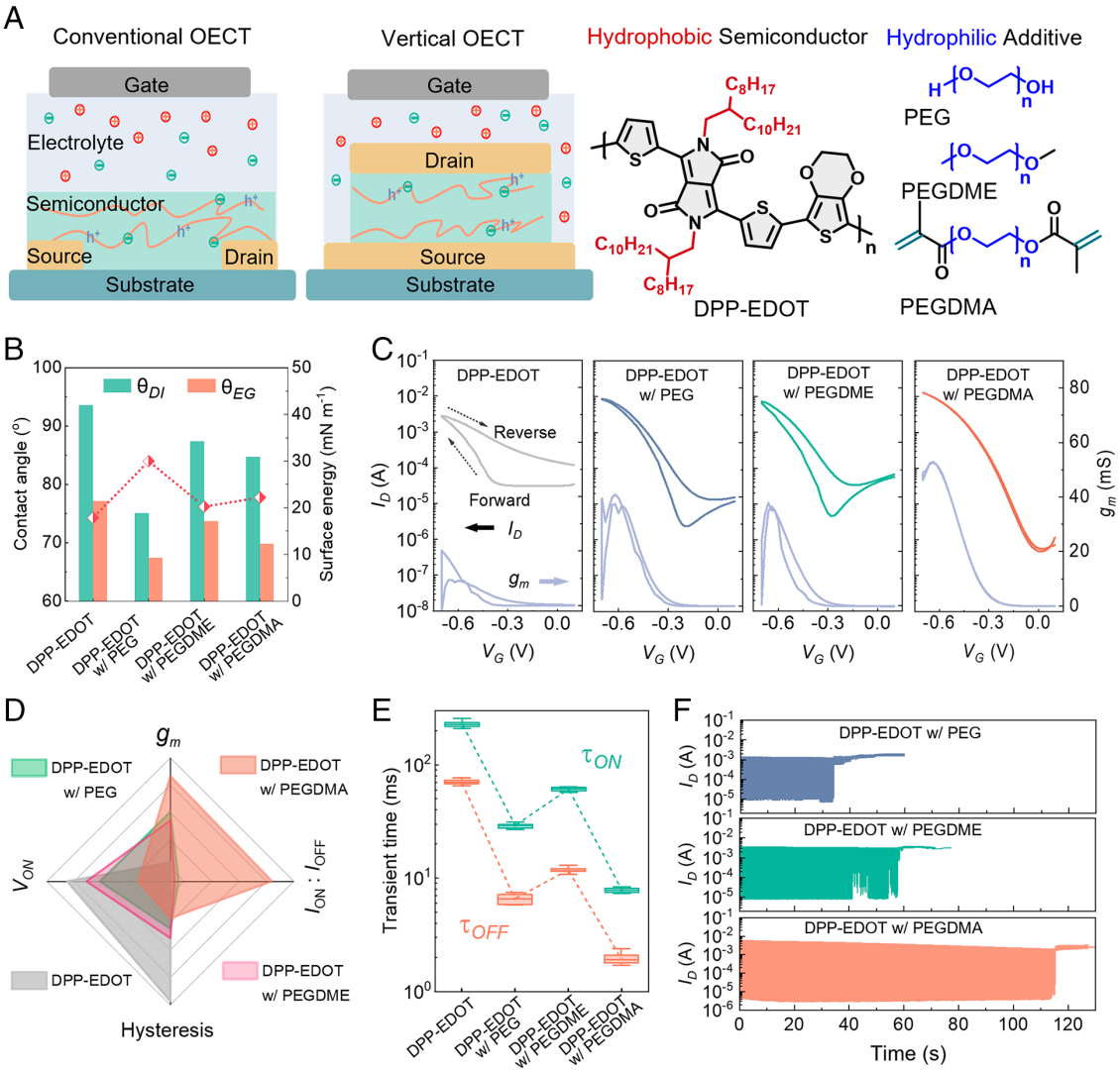



**Fig. 1.** Hydrophilic additive selection and vOECT performance. (*A*) Schematic illustration of conventional and vertical OECT architectures and chemical structures of the semiconductor DPP-EDOT and ionically conducting hydrophilic additives used in this study. (*B*) Contact angles (DI and EG) and surface energy varies with the indicated hydrophilic additives. (*C*) Transfer characteristics and the corresponding transconductance variations for vOECTs ($V_D$ = −0.5 V) using DPP-EDOT and the indicated blends including a control neat DPP-EDOT/PEG blend at a 1:0.5 mass ratio, DPP-EDOT/PEGDME blend at a 1:1 mass ratio, DPP-EDOT/PEGDMA blend at a 1:1 mass ratio. (*D*) Device performance comparison of neat and three hydrophilic additive-containing vOECTs, highlighting peak transconductance ($g_m$), turn-on voltage ($V_{ON}$), on–off ratio, and hysteresis window. (*E*) Transient time of neat and three indicated hydrophilic additive- vOECTs ($V_G$ switching between 0 V and −0.7 V, and $V_D$ = −0.5 V). (*F*) Cycling stability of three indicated hydrophilic additive-containing vOECTs. For cycling measurements, PEG- and PEGDME-containing vOECTs operated at gate voltages from 0 to −0.5 V at a 5 Hz frequency, $V_{DS}$ = −0.5 V. PEGDMA-containing vOECTs operated at gate voltages from 0 to −0.5 V at a 10 Hz frequency, $V_{DS}$ = −0.5 V. Note, all films were not crosslinked.

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Here we report a straightforward, efficient, and broadly applicable blending approach that utilizes readily available hydrophobic semiconductors to achieve high-performance OECTs. After screening a collection of possible hydrophilic additives, we report that blending photocrosslinkable poly(ethylene glycol)-dimethylacrylate (PEGDMA) into cOECT and vOECT active layers significantly enhances ion exchange rates, facilitates ionic-electronic coupling, and increases redox stability while simultaneously enabling substantial electronic charge transport for several hydrophobic semiconductors. This hydrophilic additive is broadly applicable to variety of well-known hydrophobic and amphiphilic p- and n-type semiconductors, compatible with diverse electrolytes (e.g., $KPF_6$, $KClO_4$, EMIM-TFSI), and compatible with both cOECTs and vOECTs. Thus, on introducing PEGDMA, the average OECT performance parameter increases dramatically [transconductance ($g_m$) = +244 to 133,900% (vOECT), $g_{m\ =}$ +130 to 6,771% (cOECT); on/off current ratio ($I_{ON}/I_{OFF}$) = +24,400 to 2,210,000% (vOECT) = +1,010 to 4,090% (cOECT); turn-on voltage ($V_{ON}$): = −44 to 95% (vOECT) = −19 to 41% (cOECT); hysteresis windows = −98% (vOECT); switching speed ($\tau_{ON}/\tau_{OFF}$)= −98% (vOECT)]. Furthermore, exploiting the cross-linkable structural properties of PEGDMA, scalable photopatterning is achieved for the high-resolution patterning of both semiconductors (<0.4 μm) and electrolytes (<2 μm), yielding solid-state OECTs with high performance and cycling stability (>10,000 cycles) for wafer-scale integration: 2,548 OECTs/2" wafer, and inverter, NAND and NOR logic circuits. Finally, solid-state OECT-driven photosensors are fabricated to create a 10 × 10 pixel vision sensing array mimicking the brain's complex processes of visual compression and reconstruction.

## Results and Discussion

**Hydrophilic Additive Selection and vOECT Performance.** We first identified optimal hydrophilic additives by screening different blends using the same hydrophobic polymer [diketopyrrolopyrrole-3,4-ethylenedioxythiophene (DPP-EDOT)] as a testbed and quantifying charge transport in a vOECT architecture (Fig. 1*A*, see processing and geometry details in *Materials and Methods*) since ion exchange is more challenging than in cOECTs. Three hydrophilic additives were selected: polyethylene glycol (PEG), PEG dimethyl ether (PEGDME), and PEGDMA, where the hydrophilicity slightly decreases across the series, with the latter enabling crosslinking via photopolymerization (vide infra). As expected, additive inclusion enhances hydrophilicity of the semiconductor from 17.7 mN $m^{-1}$ to ~30 mN $m^{-1}$ for PEG devices, with those incorporating PEGDME and PEGDMA being midway in between, 20.3 mN $m^{-1}$ and 22.2 mN $m^{-1}$, and close to that of the previously reported N-glycolated polymer counterpart gDPP-EDOT, 25.2 mN $m^{-1}$ (Fig. 1*B* and *SI Appendix*, Fig. S1 and *Supplementary Note* 1). Next the vOECTs were characterized with a 0.1 M NaCl electrolyte solution and an Ag/AgCl gate electrode. As expected, the neat DPP-EDOT-based vOECTs exhibit a low on-current (2.77 ± 1.08 mA) and peak $g_m$ (21.87 ± 2.57 mS) with significant hysteresis (478 ± 37 mV), indicating poor ion penetration (45), which is rectified with glycolated gDPP-EDOT where $g_m$ ~ 68 mS (*SI Appendix*, Fig. S2). Inclusion of additives at 1:1 wt% enhances vOECT performance of all devices with PEGDMA outperforming the others (Fig. 1 *C* and *D*). PEGDMA-containing vOECTs show the highest peak $g_m$ [53.42 ± 1.74 mS vs. 40.83 ± 4.18 mS (PEG) and 37.53 ± 3.59 mS (PEGDME)], reduced turn-on voltages [(+0.02 ± 0.01 V vs. −0.20 ± 0.03 V (PEG) and −0.27 ± 0.03 V (PEGDME)], and highest on/off ratio [(3.30 ± 0.52) × $10^4$]. Additionally, the PEGDMA-containing devices exhibit minimal hysteresis (9.66 ± 3.61 mV), compared to PEG (80.20 ± 14.25 mV) and PEGDME (121.42 ± 27.83 mV), reflecting enhanced ionic conductivity (Table 1). Note, vOECT performance increases are far more dramatic when using more common hydrophobic semiconductors (vide infra). In contrast, and as expected, the use of Cin-Cell dramatically reduces DPP-EDOT OECT performance (*SI Appendix*, Fig. S3), highlighting the fundamentally different function of PEGDMA vs. Cin-Cell.

vOECT transient response and cycling stability were next evaluated using the various blends. As shown in Fig. 1*E*, PEGDMA-based vOECTs exhibit a turn-on transient time ($\tau_{ON}$) of less than 8 ms and a turn-off time ($\tau_{OFF}$) of less than 2 ms, significantly faster (~10× improvement) vs. PEGDME-based vOECTs ($\tau_{ON}$ = 61.04 ms, $\tau_{OFF}$ = 11.84 ms). This improvement is ascribed to the hydrophobic PEGDME end-groups that hinder ion transport. Additionally, the on-current of neat DPP-EDOT based OECT drops by 24% after 200 consecutive doping/dedoping cycles (*SI Appendix*, Fig. S4), and the higher surface energy of the PEG additive leads to faster dissolution in aqueous electrolytes, resulting in poor cycling stability (~160 cycles, Fig. 1*F*). Note that PEGDME and PEGDMA offer better stability than PEG, with the PEGDMA-based vOECTs achieving ~1,100 cycles, likely reflecting the cross-linkable end-groups that form more stable films. Note also that stability can be further increased to over 10,000 cycles, addressing the key issue of hydrophilic additive blends dissolving in electrolytes (vide infra). These findings indicate that PEGDMA, with its polar glycol chain and functional end-groups, is the most effective hydrophilic additive for mixed ionic-electronic coupling in hydrophobic semiconductors, providing a highly effective alternative to complex side-chain engineering and without additional synthetic challenges.

**Table 1. Comparisons of vOECT performance and surface energy of various semiconductor components**

| Semicond. | SE (mN $m^{-1}$) | Peak $g_m$ (mS) | $V_{ON}$ (V) | $I_{ON}$:$I_{OFF}$ | Hysteresis (mV) | $\tau_{ON}$ (ms) | $\tau_{OFF}$ (ms) | Stability (cycles) |
|---|---|---|---|---|---|---|---|---|
| DPP-EDOT | 17.9 | 21.87 ± 2.57 | −0.38 ± 0.03 | (1.57 ± 0.71) × $10^2$ | 478.00 ± 37.01 | 229.31 ± 15.36 | 71.01 ± 3.62 | <200 |
| DPP-EDOT w/ PEG | 29.9 | 40.83 ± 4.18 | −0.20 ± 0.03 | (3.42 ± 0.19) × $10^3$ | 80.20 ± 14.25 | 29.21 ± 2.01 | 6.59 ± 0.27 | ~160 |
| DPP-EDOT w/ PEGDME | 20.3 | 37.53 ± 3.59 | −0.27 ± 0.03 | (1.58 ± 0.26) × $10^3$ | 121.42 ± 27.83 | 61.04 ± 2.38 | 11.84 ± 0.66 | ~200 |
| DPP-EDOT w/ PEGDMA | 22.2 | 53.42 ± 1.74 | +0.02 ± 0.01 | (3.30 ± 0.52) × $10^4$ | 9.66 ± 3.61 | 7.82 ± 0.36 | 1.95 ± 0.21 | ~1,100 |

Note, all vOECT data are averages of 6 vOECTs tested, and all OECTs were obtained by directly blending the semiconductor DPP-EDOT with the hydrophilic additives, without crosslinking. To quantify the hysteresis windows, voltage differences are determined by calculating the absolute drain current magnitude of $10^{-4}$ A for all vOECTs. SE represents surface energy.

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**DPP-EDOT:PEGDMA Blend Transport and Microstructure.** The film wettability, morphology, microstructure, and vOECT performance characteristics of the DPP-EDOT:PEGDMA blends were next investigated in more detail for mass ratios ranging from 10:1 to 1:2. Thus, addition of a small PEGDMA amount increases the surface energy from 17.87 mN $m^{-1}$ to ~20 mN $m^{-1}$ and further additions increase it but to a lesser extent, reaching a maximum of 22.47 mN $m^{-1}$ for the 1:2 blend (*SI Appendix*, Fig. S5 and Table S1). vOECTs were fabricated as described previously and performance details are included in *SI Appendix*, Figs. S6 and S7, Table S2, and *Supplementary Note* 2. As shown in Fig. 2*A*, as the PEGDMA content increases the OECT performance at first increases, optimizes for the 1:1 ratio, and then falls. Thus, excessive PEGDMA (1:2 ratio) reduces device performance due to the limited semiconductor mass content. Furthermore, transient measurements (*SI Appendix*, Fig. S8) indicate strong reduction of both $\tau_{ON}$ (228.4 → 7.6 ms) and $\tau_{OFF}$ (70.4 → 2.1 ms) as the PEGDMA content increases (Fig. 2*B*). Note that the incorporation of PEGDMA into the ion-permeable semiconductor gDPP-g2T further accelerates the switching dynamics, yielding $\tau_{ON}$ and $\tau_{OFF}$ of 67.5 μs and 18.1 μs, respectively—a nearly fivefold increase over our previous work (*SI Appendix*, Fig. S9) (45). Comparative benchmarking against existing OECT literature highlights the significant improvement in switching dynamics enabled by PEGDMA, even outperforming those of specially synthesized hydrophilic semiconductors (Fig. 2*C*). Thus, blending the electrically conductive semiconductor with the ionically conductive hydrophilic additive PEGDMA offers a promising direction for the future development of high-frequency OECT channel materials.

To understand the observed variations in OECT performance, the blend film microstructures were next analyzed by grazing-incidence wide-angle X-ray scattering (GIWAXS). The blending mass ratio of DPP-EDOT and PEGDMA was adjusted from 1:0 to 1:2 in accordance with the device fabrication conditions. As shown in Fig. 2*D* and *SI Appendix*, Fig. S10, all films display a strong π–π (10) diffraction peak in the out-of-plane (OOP) direction and a lamellar (100) stacking peak in the in-plane (IP) direction due to the π-face-on dominant packing orientation originating from DPP-EDOT as well as for the blend, as evidenced by the ring-like halo at 1.5 Å due to isotropic PEGDMA. This result suggests that, irrespective of the blend ratio, both hydrophobic DPP-EDOT and hydrophilic PEGDMA retain their respective microstructural characteristics. The detailed scattering parameters of these blend films are summarized in *SI Appendix*, Tables S3 and S4. Note that with increasing mass ratio of PEGDMA, the d-spacing of π–π (010) and lamellar (100) peaks gradually decrease ($d_{\pi}$= 3.72 to 3.68 Å/$d_{lam}$= 22.26 to 21.38 Å), while the crystal coherence lengths (CCLs) increase ($CCL_{\pi}$= 22.02 to 36.61 Å/$CCL_{lam}$= 69.70 to 115.44 Å). The relative degree of crystallinity (rDOC) gradually falls, correlating with greater amorphous region content that should facilitate and dominate ionic transport via greater PEGDMA densities (*SI Appendix*, Fig. S11). These results indicate that amorphous PEGDMA domains confine nano-crystals of DPP-EDOT, generating more compact and larger semiconductor crystallites (54, 55). Note that the increased π-face-on oriented crystallite content likely favors efficient charge transport in the vOECT vertical direction, while at the same time, the hydrophilic PEGDMA domains provide pathways for rapid and efficient ion transport. AFM imaging corroborates this morphological evolution (*SI Appendix*, Fig. S12), with the film rms roughness ($\sigma_{RMS}$) gradually increasing from 0.56 nm (neat DPP-EDOT film) to 1.91 nm (1:2 blend film) as the PEGDMA content increases, reflecting the phase differences (*SI Appendix*, Fig. S13).

We next investigated the morphological and structural characteristics of the neat semiconductor film as well as the 1:1 blend film, which afford optimum vOECT performance. As illustrated in *SI Appendix*, Fig. S14, the neat film exhibits a uniform topology on both the top and bottom surfaces, whereas the DPP-EDOT phase of the blend film self-assembles into a fibrillar interconnected structure. This nanoscale morphology is created by phase separation, as validated by the distribution differences in the high-resolution transmission electron microscope (TEM) images and associated energy dispersive X-ray spectroscopy (EDS) mapping of S (Fig. 2*E*). Therefore, the blend film in the horizontal direction exhibits a uniform distribution of nanoscopic semiconductor-electrolyte domains. Of particular interest is whether ionic and electronic coupling and transport occur concurrently throughout the vertical direction. Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS) depth profiling on neat and blend film samples was conducted to obtain a more comprehensive understanding of the DPP-EDOT and PEGDMA vertical spatial distribution (Fig. 2*F*). Notably, the unique signals of the semiconductor are marked by the presence of S and O atoms, while the signals associated with PEGDMA are only indicated by O atoms. Thus, PEGDMA can be identified by analyzing the relative amounts of these two elements. Enhanced O signal intensity is observed in comparison to the neat film, and the blend film shows uniform signal intensity from the top to the bottom surface, revealing that both DPP-EDOT and PEGDMA are uniformly distributed within the nanoscale vertical channel. Consequently, the uniformly distributed structures in both the horizontal and vertical directions—formed by the ionic conductor PEGDMA and electronic conductor DPP-EDOT within confined microstructures in the transistor channel are critical for ensuring the presence of highly efficient ionic-electronic coupling and transport pathways.

The volumetric capacitance (C*) of the blend films was determined by electrochemical impedance spectroscopy (EIS) (*SI Appendix*, Fig. S15 and *Supplementary Note* 3). Thus, as the PEGDMA content is increased a progressive reduction in electrochemical impedance is measured, reflecting enhanced ion penetration and transport. Correspondingly, C* follows a trend consistent with the OECT performance, with the highest-performing OECT exhibiting a 62-fold increase in C*, reflecting an increased volumetric doping-redox capacity (*SI Appendix*, Fig. S16). Next, spectroelectrochemical analysis was conducted to assess the electrochemical redox characteristics. The π–π* transition peak gradually diminishes beyond 0.5 V in the neat DPP-EDOT film, while the blend film shows a regular diminution between 0 and 0.1 V (*SI Appendix*, Fig. S17). This indicates that the oxidation potential required for electrochemical doping is significantly lower, aligning well with the turn-on voltage variations observed in the vOECT devices (Fig. 1*C*). Additionally, full recovery of the π–π* transition absorption intensity is achieved at 0 V for the blend film, in contrast to the incomplete recovery in the neat film. Note that the incorporation of hydrophilic PEGDMA increases the semiconductor's permeability to ambient $O_2$ and moisture, facilitating unintentional p-type doping from ambient $O_2$, which accounts for the enhanced optical absorption feature near 1,300 nm in the blend film. These differences in stable, reproducible electrochromic behavior argue that incorporating PEGDMA into the hydrophobic film enhances the efficiency of ion doping/dedoping in the electrochemical process.

**Generality of the Blend Strategy for v/cOECTs.** We next investigated whether PEGDMA is compatible with other electrolytes and capable of promoting OECT charge transport in other hydrophobic semiconductors. To explore this, $ClO_4^-$, $PF_6^-$ (aqueous electrolytes), and $TFSI^-$ (an ionic liquid) ions were

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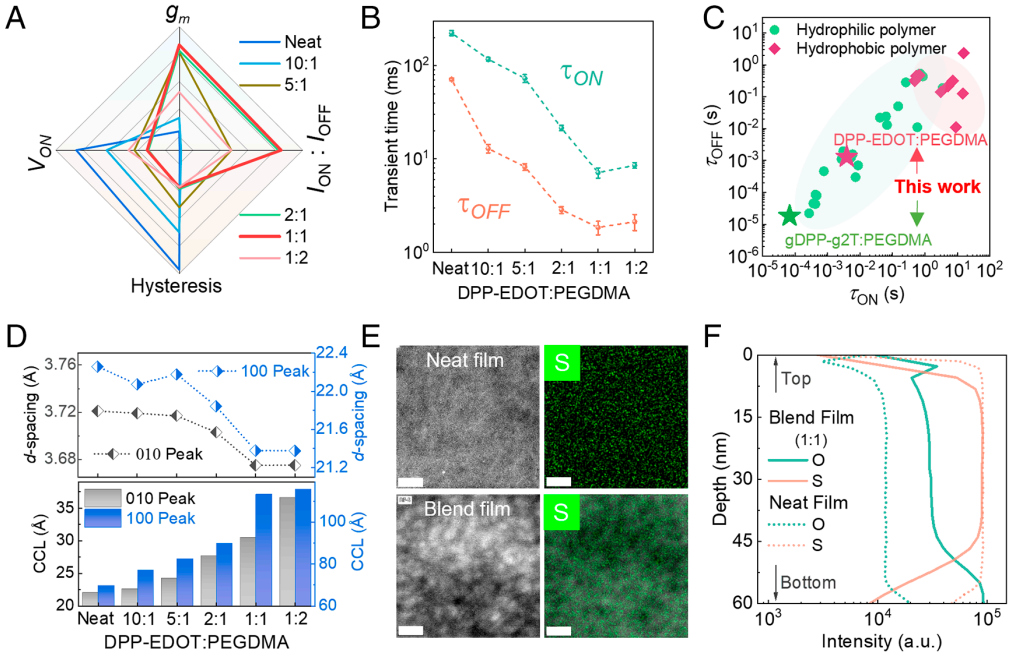



**Fig. 2.** Effect on OECT performance, microstructure, and operating mechanism of DPP-EDOT:PEGDMA blend composition. (*A*) Correlation of device parameter variations with different DPP-EDOT and PEGDMA blend ratios, including $g_m$, $V_{ON}$, hysteresis window, and $I_{ON}$:$I_{OFF}$. (*B*) Transient times variation with different DPP-EDOT and PEGDMA blend ratios. (*C*) Transient time comparison of the present OECTs with published results for both hydrophilic and hydrophobic semiconducting materials (53). (*D*) Comparison of film d-spacing and CCL values with various blend mass ratios of DPP-EDOT and PEGDMA. (*E*) High-resolution TEM images of neat DPP-EDOT and blend for DPP-EDOT:PEGDMA =1:1 films. (Scale bar, 100 nm.) (*F*) ToF-SIMS derived composition distribution of S and O elements from the top to the bottom surface of neat DPP-EDOT and blend films.

used as electrolytes for OECTs based on the DPP-EDOT blend (*SI Appendix*, Fig. S18). All key OECT parameter trends are consistent with those observed using chloride ions (*SI Appendix*, Table S5). Despite the very different intercalation tendencies of these anions (43), the minor parameter differences observed suggest that ion transport is dominated by PEGDMA incorporation. To further evaluate the broad applicability of this blend strategy, we fabricated vOECTs using a range of hydrophobic polymeric and molecular p- and n-type semiconductors, including P3HT, PBTTT-C14, DPP-TT, N2200, PYIT, and Y6 (Fig. 3*A*). Contact angle measurements confirm that all blends become significantly more hydrophilic, with surface energies ranging from 16.0 to 20.3 mN $m^{-1}$, compared to those of the neat films, which ranged from 21.6 to 30.2 mN $m^{-1}$ (*SI Appendix*, Fig. S19). AFM surface morphology analysis further confirms that incorporating PEGDMA uniformly increases the surface roughness of the films (*SI Appendix*, Fig. S20). All blend-based vOECTs outperform their neat semiconductor counterparts, with p-type devices exhibiting high current levels (>10 mA), large on/off ratios (>$10^5$), low oxidation potentials (<0.3 V), and negligible hysteresis (Fig. 3*B*). Notably, more challenging n-type semiconductors, such as N2200, PYIT, and Y6, also perform well in blends. These devices display robust, reversible transfer curves, low reduction potentials (<0.4 V), relatively high current (>0.5 mA), and high on/off ratios (>$10^3$) (Fig. 3*C*). Furthermore, PEGDMA addition improves transport properties of amphiphilic and highly textured hydrophilic semiconductors such as DPP-g2T and Pg2T-T, respectively (Fig. 3*D*) (32, 56).

Finally, this strategy is extendable to OECTs with a conventional architecture (cOECTs), thus facilitating ion exchange at both μm and/or nm interfacial scales (*SI Appendix*, Fig. S21). Fig. 3*E* and *SI Appendix*, *Supplementary Note 4* summarize the volume-normalized peak $g_m$ ($g_{m,n}$) for all semiconductors studied. Thus $g_{m,n}$ of p- and n-type cOECTs with PEGDMA reaches 63.33 S $cm^{-1}$ and 0.07 S $cm^{-1}$, respectively, vs. 0.28 S $cm^{-1}$ and inactive state for the neat semiconductors, respectively. For vOECTs, the commonly used area-normalized peak $g_m$ ($g_{m,A}$) achieves values exceeding 335.00 μS $μm^{-2}$ for p-type blends and 6.08 μS $μm^{-2}$ for n-type blends vs. negligible numbers for the neat semiconductors. These results clearly demonstrate that the vOECTs exhibit significantly enhanced performance, achieving area-normalized transconductance values markedly superior to those reported previously (*SI Appendix*, Fig. S22). Consequently, this blend strategy is broadly applicable in terms of semiconductor material type (p-, n-type, hydrophobic, amphiphilic, and hydrophilic), electrolyte type (commonly used aqueous and ionic liquid) as well as device architecture type (vOECT and cOECT).

**Photo-Crosslink Patterning and Wafer-Scale Integration.** To leverage the crosslinkable acrylic units in PEGDMA, we developed a direct OECT photopatterning technique using free radical polymerization promoted by the phenylbis(2,4,6-trimethylbenzoyl)phosphine oxide (BAPO) radical initiator (see crosslinked network structure in Fig. 4*A* and details in *SI Appendix*, Fig. S23). The formation of this network is confirmed by FTIR spectroscopy (*SI Appendix*, Fig. S24), where the characteristic methacrylate C=C bands at ≈1,610 $cm^{-1}$ disappear under irradiation. Unlike earlier crosslinking strategies using azide-based crosslinking reactions limited to polymeric semiconductors (56, 57), the present approach enables direct patterning of both the semiconductors and ion gel electrolytes (Fig. 4*B*). This photocrosslinking modulates the solubility of deposited semiconductor and/or electrolyte films, making the UV-irradiated (365 nm) regions resistant to solvent development, while the non-UV-irradiated regions are rapidly removed during the development process. Accordingly, predesigned square patterns with feature sizes of 200 × 200 $μm^2$ for semiconductors (DPP-EDOT/PEGDMA blend) and ionic gel electrolytes (EMIM-TFSI/PEGDMA blend) are obtained (Fig. 4 *C* and *D*). Furthermore, the patterning resolution can be scaled down to 0.4 μm for semiconductors and

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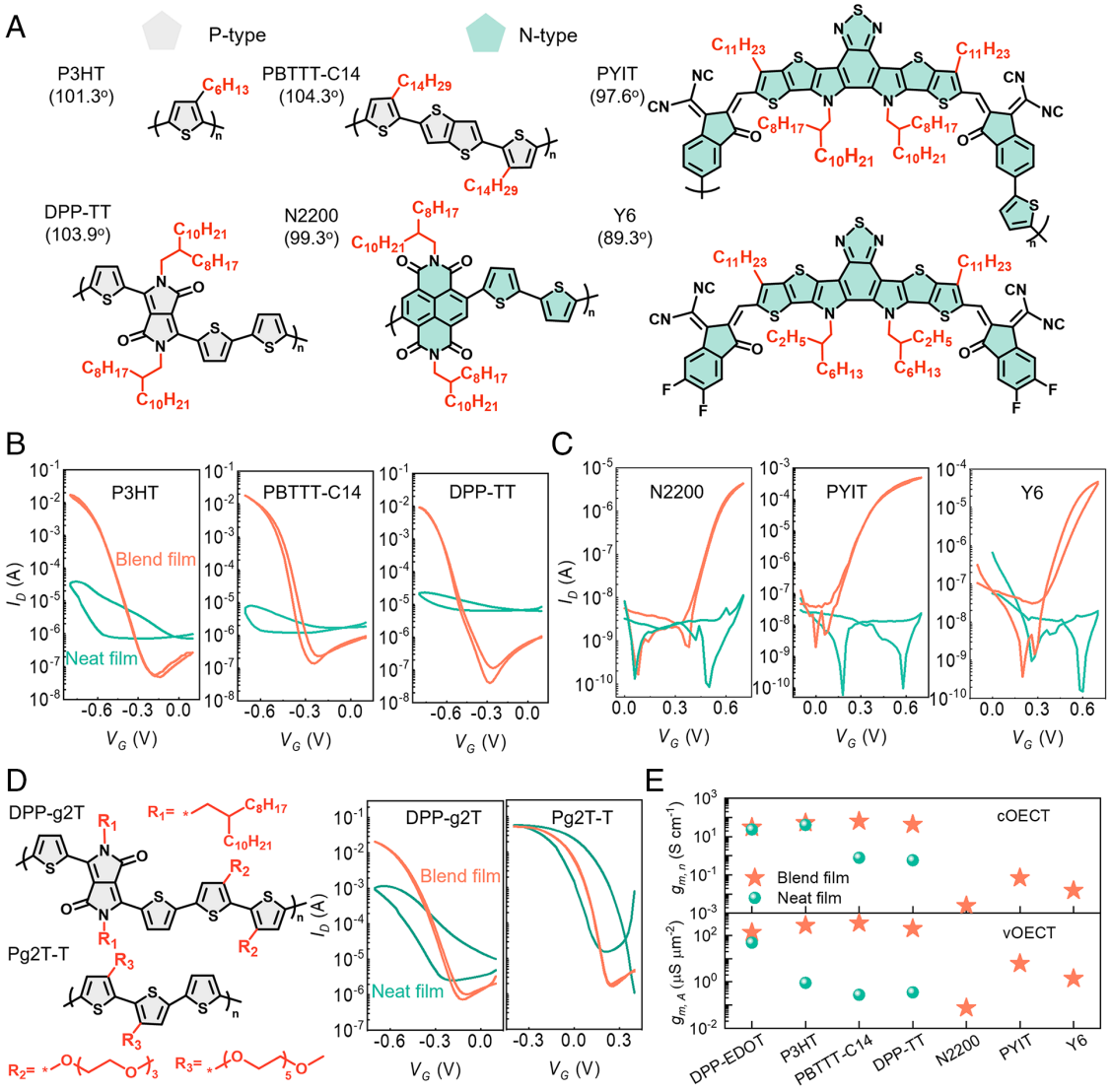



**Fig. 3.** Broad generality of the blend approach for v/cOECTs. (*A*) Chemical structures of conventional p- and n-type hydrophobic semiconductors (polymer and small molecule, including P3HT, PBTTT-C14, DPP-TT, N2200, PYIT, and Y6). Values in parentheses are the contact angle with DI. (*B*) Generality of p-type semiconductors, including neat and blend films based on vOECTs, where the electrochemical performance is improved by two to three orders of magnitude. (*C*) Generality of n-type semiconductors, including neat and blend films based on vOECTs. For n-type polymer or small molecule OECTs, the transition from inactive electrochemical performance to a high-performance accumulation mode of operation is observed. Here, all the p- and n-type vOECTs were characterized with a 0.1 M NaCl electrolyte. (*D*) Generality of the hydrophilic PEGDMA additive in enhancing OECT performance for selected OMIEC materials exhibiting poor performance. Chemical structures of organic semiconductors used for OECTs, DPP-g2T, and Pg2T-T. (*E*) Normalized peak transconductance summary for vOECTs and cOECTs. Note that vOECTs use area-normalized transconductance, while cOECTs use volume-normalized transconductance.

2 μm for electrolytes (*SI Appendix*, Fig. S25 and *Supplementary Note* 5), representing the highest resolution reported to date for patterned organic semiconductors and electrolytes. Thus, hydrophilic PEGDMA demonstrates multifunctionality beyond simply enhancing ionic-electronic coupling and transport in diverse semiconductors (as discussed earlier).

As a further example, the transport properties of DPP-EDOT vOECTs using a 0.1 M NaCl aqueous solution as electrolyte, remain unaffected or even improved after the photo-patterning of the transistor channel. This is likely due to reduced parasitic capacitance, as evidenced by the ~10× decreased off current, while other physical parameters remain essentially unchanged, as evident in the transfer (Fig. 4*E*) and transient (*SI Appendix*, Fig. S26) responses. Notably, this semiconductor photo-patterning approach can also be extended to other high-performance p- and n-type OMIEC materials (*SI Appendix*, Fig. S27). In addition to semiconductor patterning, we fabricated all-photopatterned solid-state OECTs (APSS-OECTs). This approach minimizes unwanted ionic migration and electrochemical side reactions. The representative APSS-OECT transfer curves indicate typical p-type characteristics, and lower off-state currents were observed after electrolyte patterning (Fig. 4*F*). More importantly, incorporating a crosslinked nanoscopic channel through hydrophilic PEGDMA further stabilizes the microstructure of the as-cast semiconductor films, enhancing resilience to electrolyte-induced swelling and reducing channel degradation accompanying hydrated ion injection/ejection processes (45, 56, 58). As a result, both crosslinked liquid-state and solid-state vOECTs exhibit a 10× greater cycling stability vs. noncrosslinked vOECTs, with stability exceeding 10,000 cycles in both cases (Fig. 4*G*). Moreover, extended stability measurements at 0.5 Hz were performed to highlight the device's long-term operational stability, accumulating a total test duration exceeding 120 min (*SI Appendix*, Fig. S28).

Note that the ability to fabricate high-performance APSS-OECTs enables high-resolution circuit design, facilitating integration into

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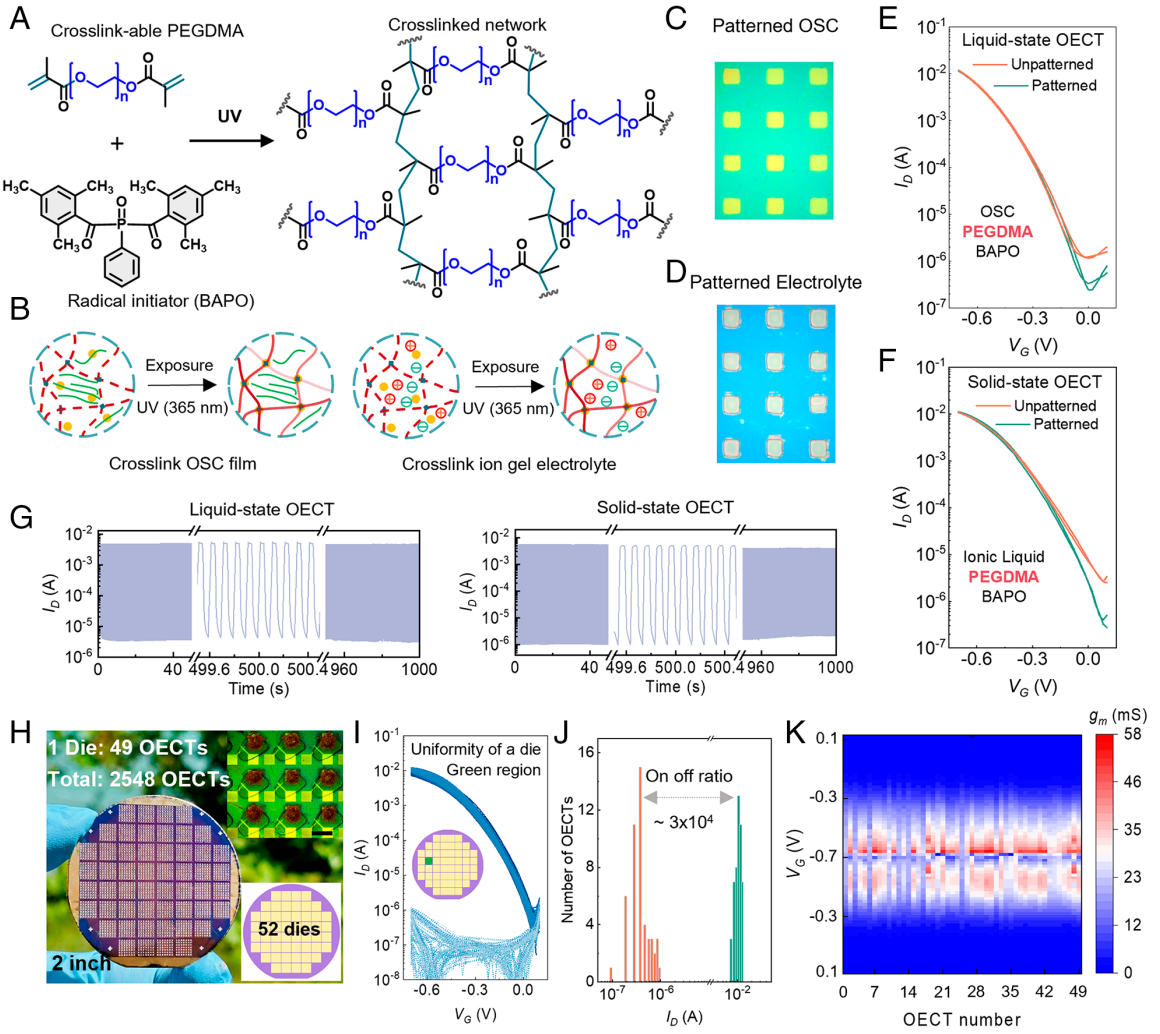



**Fig. 4.** Photo-crosslinking for patterning and wafer-scale integration. (*A*) Crosslinking mechanism of PEGDMA using the photoradical initiator BAPO. (*B*) Schematic illustration of the photo-crosslinking process for organic semiconductors and ion gel electrolytes using PEGDMA and BAPO. (*C*) Optical image of a representative photopatterned DPP-EDOT square with a 200 μm resolution. (*D*) Optical image of a representative photopatterned ion gel electrolyte square with a 200 μm resolution. (*E*) Representative transfer curves of a vOECT operating with 0.1 M NaCl electrolyte and unpatterned or patterned semiconductors. Note, the transistor channels of both OECTs are composed of DPP-EDOT, PEGDMA, and BAPO in the same mass ratio, and $V_D$ = −0.5 V. (*F*) Representative transfer curves of a vOECT operating with a solid electrolyte and unpatterned or patterned ion gels. Note, the electrolyte is composed of EMIM-TFSI ionic liquid, PEGDMA, and BAPO, the semiconductors for OECTs are patterned, and $V_D$ = −0.5 V. (*G*) Cycling stability of a vOECT (DPP-EDOT semiconductor) for liquid-state and solid-state crosslinked OECTs, allowing operation at >10,000 cycles. For cycling stability measurement, gate pulse changed from 0 to −0.5 V with 10 Hz frequency, $V_D$ = −0.5 V. (*H*) Photograph of a 2-inch wafer-scale integration of all patterned solid-state OECTs, having 52 dies within 2548 OECTs. Zoomed inserted microscope image shows details of the solid-state OECT. (Scale bar, 400 μm.) (*I*) Transfer characteristics of the solid-state OECTs in a die for 49 OECTs with a yield of 100%; the highlighted green region shows distribution area of the measured OECTs. (*J*) Histogram showing the distribution of on-current and off-current obtained from 49 solid-state OECTs in a die. The data illustrate the variability in on/off current across the measured OECTs. (*K*) Map plot showing the variation of 49 OECT transconductances on one die. The transconductance value for each OECT is the distribution of all measured voltages (both forward and reverse sweep direction) for that device along the y axis.

wafer-scale arrays and miniaturized circuits. To explore uniformity and scalability, we fabricated a 2,548-device OECT array (52 × 49, 52 dies each with 49 OECTs) on a 2" wafer, with each device featuring a 20 μm-wide and deep channel (Fig. 4*H*, and *SI Appendix*, Fig. S29). The robust fabrication of each layer (Au source/drain, patterned semiconductor, Ag/AgCl gate, and electrolyte) ensures consistent electrical performance, yielding 100% uniformity and p-type behavior across all devices (Fig. 4*I*), with a drain current on–off ratio >$3.0 \times 10^4$ and consistent transconductance (Fig. 4 *J* and *K*). Testing devices across the wafer confirmed scalability, although larger areas show some nonuniformity in current and transconductance, while maintaining on–off ratios ~$2.0 \times 10^4$ (*SI Appendix*, Fig. S30). We next developed APSS-OECT logic circuits, including inverters and logic gates, which demonstrate the expected functions (*SI Appendix*, Figs. S31–S34 and *Supplementary Notes* 6 and 7), thereby overcoming challenges in liquid-gate OECTs and enabling high-density integration for sensing, storage, and logic operations.

**APSS-OECT Based Vision Sensor Array for Compressed Sensing.** As a proof of concept, we next designed and fabricated a vision sensing array using APSS-OECTs, which enable individual vision pixel control and facilitate complex compression and reconstruction of visual images, akin to the brain's visual perception. As shown in Fig. 5*A*, the human visual system efficiently processes sparse samples from the retina, compressing and reconstructing visual signals in the brain. To mimic this process, the present vision array performs sparse sampling to compress images into low-dimensional signals, which, combined with deep learning, reconstruct high-dimensional images with features similar to the originals. Our 10 × 10 APSS-OECT pixel array integrates solid-state OECTs, organic resistors, and organic photosensors using only photopatterning

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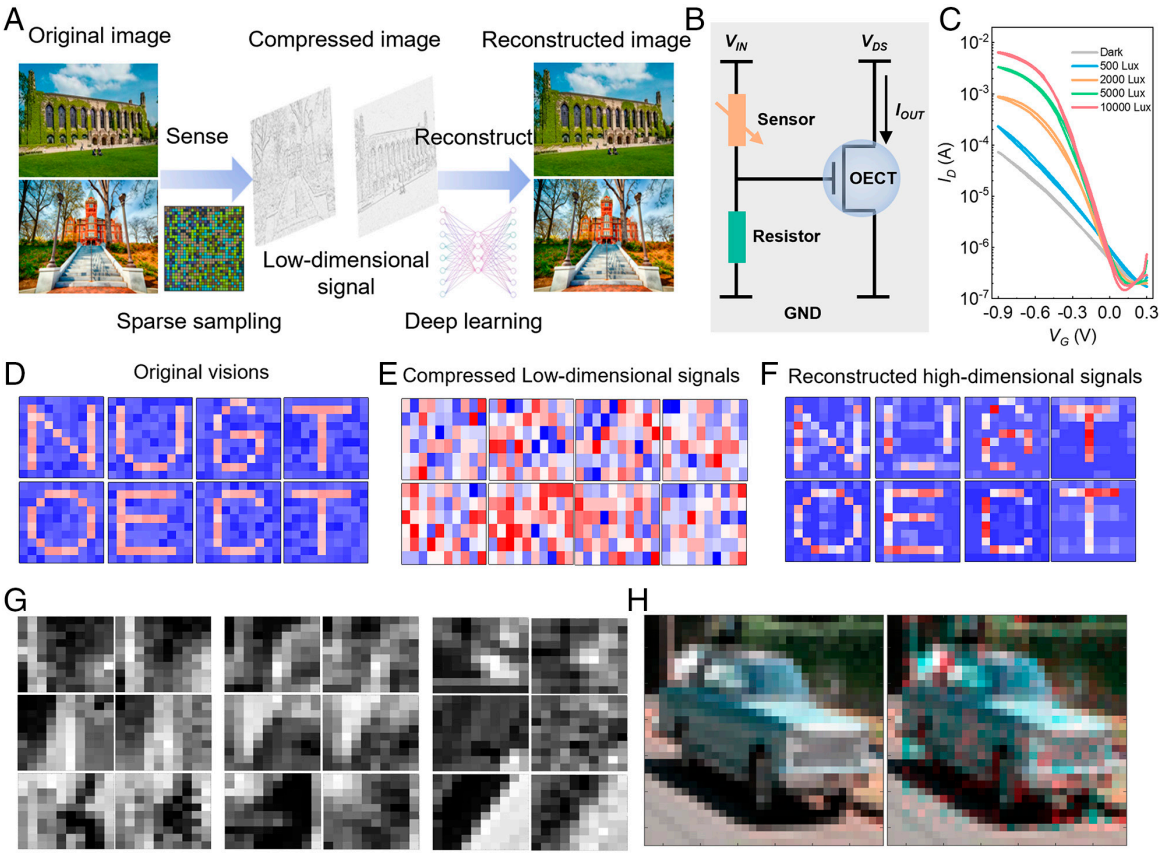



**Fig. 5.** All patterned solid-state OECT array-based vision sensor for compressed sensing. (*A*) Schematic illustration of the vision sensing system for real-image sensations from retina to brain, and the process flow of the all solid-state OECT array based compressed sensing system, emulating the human vision system for image signal processing with sensing, compression and reconstruction process. (*B*) Schematic of an integrated photo-responsive circuit unit with signal amplification capabilities. (*C*) Transfer characteristics of the photo-response of the integrated unit sensory element in the dark state and under varying light intensities, $V_{DS}$ = −0.5 V. Owing to the variability in the relative resistances of the resistor and the photosensor, the characteristic output curve can be modulated by the OECT by up to two orders of magnitude. (*D*) Original vision information captured via fabricated OECT vision sensation array. (*E*) Vision information captured by a sparse sampling rate of 60% with fabricated OECT vision sensation array, indicating the compressed low-dimensional signal. (*F*) Visual information reconstructed after signal processing, even if the sampling rate is only 60%. (*G*) A demonstration of more complexed image compression examples using the 10 × 10 array with a sampling rate of 50%, where columns 1, 3, and 5 represent the original images and columns 2, 4, and 6 represent the reconstructed images. (*H*) Higher density and complicated image compression demonstration (30 × 30 array) for comparing the original truck with the image after compression and reconstruction with 50% sampling rate. *Left*, original image, *Right*, reconstructed image.

techniques (*SI Appendix*, Fig. S35). Fabrication details are provided in *Materials and Methods* and *SI Appendix*, Fig. S36. In each pixel, the light-modulated optoelectronic circuit acts as a photovoltaic divider, distributing potential to the OECT's gate, based on the resistance ratio between the photosensor and load resistor (Fig. 5*B*). Testing reveals a load resistor current of −1.62 nA at −0.9 V, while photosensor current varies from −0.76 nA to −4.67 nA, allowing precise light-based control of gate potential for amplifying visual signals (*SI Appendix*, Fig. S37).

Upon illuminating the vision sensor with light intensities from 0 to 10,000 lx, the on-current ($V_{IN}$ = −0.9 V, $V_{DS}$ = −0.5 V) of the access OECT increases from < 0.1 mA to ~10 mA (Fig. 5*C*), with good voltage-controlled switching (*SI Appendix*, Fig. S38). This high current response, modulated by optical intensity, enables precise mapping of input images in high-dimensional sampling. For example, in a 10 × 10 array, we achieve high-resolution mapping of letters like "NUGT" and "OECT" with 100-dimensional sampling (Fig. 5*D*). Using compressed sensing at a 60% sampling rate, these letters were encoded into low-dimensional signals (10 × 6 pixels), which were accurately reconstructed into 10 × 10 images (Fig. 5 *E* and *F*). Importantly, even at a reduced 10% sampling rate, key image features are preserved (*SI Appendix*, Fig. S39), with the reconstruction accuracy achieved through the artificial neural network remaining as high as 60% (*SI Appendix*, Fig. S40). To demonstrate scalability, we applied this method to a 30 × 30 pixel array, compressing it into 9 10 × 10 segments. After 50% sampling, the reconstructed 30 × 30 image retains visual consistency (Fig. 5 *G* and *H*). Thus, these APSS-OECT-based vision sensors mimic human visual perception by reconstructing high-resolution images with fewer measurements, improving energy efficiency, and making them ideal for compact, low-power imaging platforms.

## Conclusions

In this work, we have achieved significant advances in organic electrochemical transistor materials, devices, and systems by incorporating readily accessible hydrophilic additives. This simple, scalable, and efficient blending approach addresses the long-standing issue of poor ion conduction in most p- and n-type hydrophobic and amphiphilic semiconductors, ensuring compatibility with various electrolytes and device architectures for enhanced OECT performance. The use of hydrophilic PEGDMA, with its photo-cross-linkable end groups, enables high-resolution patterning of both semiconductors and electrolytes, facilitates the construction of stable liquid- and solid-state OECTs. These results should facilitate the development of solid-state logic circuits and wafer-scale OECT integration. By bridging material and device innovations,

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this approach enables high-density functional OECT and sensor integration, exemplified by an OECT-enabled vision sensing system that mimics the human eye's compression sensing capabilities. This work paves the way for applications in biomimetic sensing and neuromorphic electronics.

## Materials and Methods

**Materials.** Materials sources: Hydrophilic additives including PEG (Mw ~ 600), PEGDME (Mw ~ 500), and PEGDMA (Mw ~ 750), hydrophobic semiconductors such as P3HT (Mw ~50,000 to 100,000) and PBTTT-C14, as well as Ag/AgCl paste were obtained from Sigma-Aldrich. DPP-TT and Y6 were sourced from 1-Materials. N2200 and P(g2T-T) were acquired from Derthon Optoelectronics Materials Science Technology Co LTD, while PYIT was purchased from Solamer. DPP-EDOT, DPP-g2T, gDPP-EDOT, gDPP-g2T, Homo-gDPP, Homo-gDPPTz, Cin-Cell, and double-end functionalized trifluoromethyl-substituted diazirine (DtFDA) crosslinker were synthesized according to previously established protocols (32, 57, 59, 60). Processing solvents, including chloroform, hexane, acetone, ethylene glycol, and isopropanol, as well as salts (NaCl, $KPF_6$, $KClO_4$) for electrolytes, and the ionic liquid EMIM-TFSI, were purchased from Sigma-Aldrich. The thermal initiator of AIBN and the photo-initiator BAPO were also purchased from Sigma-Aldrich. All materials were used as received without further purification.

Precursor solution for OECTs: All semiconductors (DPP-EDOT, P3HT, PBTTT-C14, DPP-TT, N2200, PYIT, Y6, DPP-g2T, P(g2T-T), gDPP-EDOT, gDPP-g2T, and Homo-gDPPTz) were dissolved in chloroform at a concentration of 10 mg/mL. Hydrophilic additives (PEG, PEGDME, and PEGDMA) were similarly dissolved in chloroform, but at a higher concentration of 40 mg/mL, while the thermal and photo initiators were dissolved at 4 mg/mL. All solutions were stirred overnight at 600 rpm. Subsequently, the semiconductor solutions were combined with the hydrophilic additive solutions to achieve the desired mass ratios for OECT fabrication. Each blend precursor was carefully prepared and stirred for an additional 30 min before spin-coating. For demonstrating universality, the mass ratio of p-type semiconductors to PEGDMA was maintained at optimal 1:1, while that of n-type semiconductors was maintained at 2:1, with 2 wt% thermal initiator added relative to PEGDMA. For photopatterning, after adjusting the mass ratio of semiconductor to PEGDMA, the photo-initiator BAPO was incorporated at 10 wt% relative to PEGDMA. To prepare the solid electrolyte, the components were mixed according to the mass ratio EMIM-TFSI:PEGDMA:BAPO = 7:2:1, with acetone solvent added to adjust the PEGDMA concentration to 200 mg/mL.

Precursor solution for vision sensor array: The crosslinker DtFDA was first dissolved in chloroform at a concentration of 20 mg/mL, photoactive layer of P3HT and N2200 were dissolved in chloroform at a concentration of 10 mg/mL, and then were mixed as 2:2:1 (P3HT:N2200:DtFDA) in mass ratio. Homo-gDPP, used as resistor, was dissolved in chloroform at a concentration of 10 mg/mL and mixed with DtFDA in a mass ratio of 3:1. Cin Cell, used as electrical insulation layer, was dissolved in chloroform at a concentration of 20 mg/mL and mixed with DtFDA in a mass ratio of 3:1.

**Device Fabrication.** Fabrication of OECTs: vOECTs were fabricated on precleaned Si/100 nm $SiO_2$ wafers. The bottom source electrode was patterned using a shadow mask, followed by the sequential deposition of 3 nm Cr (at a controlled rate of 0.1 to 0.2 Å/s) and 100 nm Au (at 1.0 to 2.0 Å/s) via thermal evaporation. The nanoscale transistor channel was then uniformly deposited by spin-coating at 3,000 rpm for 20 s under ambient conditions. For n-type OECTs (N2200, PYIT, and Y6), a thermal initiator (AIBN) was required for optimal performance, followed by heating the film at 70 °C for 15 min. To photo-pattern the semiconductors, 10 wt.% BAPO photoinitiator relative to PEGDMA was introduced, and designated regions were exposed to UV light (365 nm) through ELC-500 UV curing chamber for 60 s in a glovebox, then developed in chloroform for 1 s and dried with nitrogen. The top drain electrode was then deposited by thermal evaporation of 100 nm Au (at 1.0 to 2.0 Å/s) using a shadow mask, with the substrate temperature maintained at ~20 °C using a back water-cooling stage to minimize thermal effects. Finally, a droplet of 0.1 M NaCl (1 μL) was introduced to the electrode overlap region. The Ag/AgCl electrode, serving as the gate, was then carefully inserted into the droplet to drive the liquid OECT. For solid-state OECTs, the liquid electrolyte and Ag/AgCl electrode were replaced with a solid-state gel electrolyte and Ag/AgCl paste. The Ag/AgCl paste was first dropped as a side gate on the predesigned region near the semiconductor and then dried at 60 °C for 20 min. The preconfigured electrolyte solution (EMIM-TFSI:PEGDMA:BAPO) was next spin-coated at 1,000 rpm for 60 s, then carefully aligned using a stationary mask to minimize damage to the electrolyte. The sample was subsequently transferred to a glovebox, exposed to 365 nm UV light for 1 min, and then sequentially developed with deionized water and hexane to the obtain patterned electrolyte, followed by thermal treatment at 60 °C for 10 min to remove residual solvents. Note, cOECT fabrication producer is described in *SI Appendix, Supplementary Note* 8.

Fabrication of all solid-state wafer-scale arrays: The two-inch wafers were first cleaned with isopropyl alcohol, then the bottom electrodes were prepared using a photolithography process to ensure overall uniformity. A layer of nLOF 2035 negative photoresist was spin-coated (3,000 rpm, 40 s) onto the wafer and baked (110 °C, 60 s), then the photoresist was exposed to 375 nm UV light utilizing a maskless aligner, followed by a postbaking treatment (110 °C, 60 s), and development using AZ 300 K for 45 s. Next, 3 nm Cr and 100 nm Au were thermally evaporated, and immersed in an acetone overnight liftoff to achieve the formation of the bottom source electrodes. Next, the semiconductors (DPP-EDOT, PEGDMA, and BAPO) were spin-coated (3,000 rpm, 20 s) and patterned using the same producers as for the vOECT above, the top drain electrodes were then thermally evaporated (100 nm) using a designed mask for wafer-scale arrays, and then the Ag/AgCl was screen printed (with dimensions of 200 × 200 $\mu m^2$) to act as a side gate. Last, the ion gel electrolyte was spin-coated on and the patterned arrays can be obtained through the same patterning process described before.

***Fabrication of APSS-OECT based vision sensor array.*** The following sequential steps were performed to fabricate an APSS-OECT-based vision sensor array, as also indicated in *SI Appendix*, Fig. S36: 1) Thermal evaporation of the source and gate electrodes (3 nm Cr and 100 nm Au) for the 10 × 10 array, with a line width of 20 μm. 2) Spin-coating of the semiconductor film (DPP-EDOT, PEGDMA, and BAPO blends) followed by patterning the OSC array using standard photolithography techniques. 3) Deposition of an electrode isolation layer to prevent short-circuiting between the vertically stacked bottom gate and top drain electrodes. A blend solution of Cin-Cell and DtFDA was applied via spin-coating at 3,000 rpm for 20 s in a glovebox, crosslinked by exposure to 365 nm UV light (450 mW $cm^{-2}$) for 2 min using a shadow mask, developed in chloroform for 3 s, and dried with nitrogen. 4) Deposition of an organic resistor layer for the voltage divider block. A blend solution of Homo-gDPP and DtFDA was spin-coated at 3,000 rpm for 20 s, crosslinked under 365 nm UV light for 3 min, and developed following the same process. 5) Deposition of the photoactive layer to construct the photosensor and control the divided voltage for driving each individual OECT. A blend solution of P3HT, N2200, and DtFDA was spin-coated at 3,000 rpm for 20 s, crosslinked under 365 nm UV light for 3 min, and developed using the same process to obtain the sensing array. 6) Thermal evaporation of the drain electrode (100 nm Au) with a linewidth of 20 μm. 7) Screen printing of 10 × 10 Ag/AgCl electrodes onto a predesigned Au pad as side gates to drive each vision pixel, followed by curing at 60 °C for 20 min, with gate electrodes of 200 × 200 $\mu m^2$. 8) Patterning of the ion gel using an established electrolyte patterning technique to ensure direct contact between the gate and semiconductor on each vision pixel.

**Device and Material Characterization.**

***Device characterization.*** All the electrical characterization of the OECTs (transfer and output curves) and circuits were conducted utilizing a probe station connected to an Agilent B1500A semiconductor parameter analyzer under ambient conditions. For OECT characterizations, a voltage sweeping speed of 0.1 V/s was employed, tungsten probes were in direct contact with the Au pads (source and drain) and/or the Ag/AgCl-printed gates to measure the data. For transient response analysis of the transistor and stability measurements, voltage pulses were meticulously generated utilizing a Keysight waveform generator (33500B), while the current-time variation was precisely monitored leveraging an Agilent B1500A instrument. For the circuit characterizations, the same waveform generator and semiconductor analyzer equipment were used to provide the input and collect the output signals.

***Materials characterization.*** Contact angles were measured by VCA Optima XE. Tapping mode AFM was conducted using a Bruker ICON System. Film thickness was conducted by Bruker ContourGT Optical Profiler-3D Optical Microscope. Depth-profiling Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS) analysis was performed using latest generation of a high-end·M6 instrument

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developed by IONTOF. TEM characterizations were carried out employing a JEOL ARM300F. Spectroelectrochemical spectra were acquired using a Cary 100 Bio UV-Visible Spectrophotometer connected with a BioLogicSP-150 potentiostat controlled by EC-Labs software. GIWAXS data were acquired at the PLS-II 9A beamline of the Pohang Accelerator Laboratory in Korea. EIS was conducted using a three-electrode configuration with an electrochemical analyzer (CHI660E). Prior to these measurements, all samples were prepared by spin-coating onto $Si/SiO_2$ substrates coated with an 8 nm gold. The dimensions of the substrates utilized were 10 mm × 10 mm, while the thicknesses of both neat and blend films were consistently maintained at approximately 60 nm.

**Simulations of In-Sensor Compression.** Current signals collected by the APSS-OECT vision sensing array were employed for in-sensor image compression (61, 62). Note, to meet the requirement of stochasticity of vision sensors during the sampling producer, we modeled the output currents of transistors with Gaussian noise $N(\mu, \sigma^2)$, representing the parameters of the expectation ($\mu$) and variance ($\sigma^2$) of Gaussian distribution. First, the original input signal X was converted into a one-dimensional column vector (N × 1) that corresponds to N pixels, which can be mapped by the vison sensor array. Second, the transistors were switched randomly (setting the input voltage pulses to make the target sensor turn on and/or off) to form a pseudorandom 0/1 pattern, which was converted into a one-dimensional row vector $\phi$ (1 × N). Third, by employing Kirchhoff's current law and Ohm's law, all the sampling currents (including on-current and off-current) of the vision sensor could be formed and accumulated to obtain the measurement vector $Y_i$ of, and this process is repeated M times to form a measurement matrix $\phi$ (M × N). The Y was compressed signal with the dimensionality of M, with reduced dimensional data from N to M (M < N) determined by the sampling rate. Fourth, the Y vector carrying all the information of the input signal X needed to be transmitted to the terminal, where reconstruction and visualization could be performed. The details on the compression and reconstruction are also described in *SI Appendix, Supplementary Note 9*. For the accuracy of artificial neural network, we use a Resnet 18, SGD optimizer, batch size 128, adaptive learning rate, and 20 iterations to recognize images with different sampling rates (60%, 50%, 40%, 30%, 20%, 10%).

**Data, Materials, and Software Availability.** Study data are included in the article and/or *SI Appendix*.

**ACKNOWLEDGMENTS.** T.J.M. and A.F. acknowledge support of this research by AFOSR contract FA9550-22-1-0423, and with J.R., by the Northwestern University Materials Research Science and Engineering Center Award NSF DMR-2308691. A.F. acknowledges US-Taiwan AFOSR grant # FA2386-24-1-4040. J.Y. thanks the NSFC (U21A20492), the Sichuan Science and Technology Program (2024YFHZ0354) and the Sichuan Province Key Laboratory of Display Science and Technology for support of this work. This work is also partially sponsored by UESTC scholarship. We thank the IMSERC NMR facility at Northwestern University, which received support from the Soft and Hybrid Nanotechnology Experimental (SHyNE) Resource (NSF ECCS-2025633), International Institute of Nanotechnology (IIN), and Northwestern University. This work also made use of the SPID facility of Northwestern University's NUANCE Center, which has received support from the SHyNE Resource (NSF ECCS-2025633), the IIN, and Northwestern's MRSEC program (NSF DMR-1720139).

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Reviewers: D.G., University of Washington; H.E.K., Johns Hopkins University; and T.-Q.N., University of California Santa Barbara.

The authors declare no competing interest.

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